

Table 1: Zero-day logic discrepancies in cross-layer communications. Here, * denotes an arbitrary data type or page element.

SuperApp	Issue	Discrepancies			Data Channel	Data Type	Page Element	Page Position	Observation Oracle	Combination	UI Interaction	Root Cause	Fixed	O ⁻	C ⁻	U ⁻
		 	 	 												
	C1	GW : EW : GS	EW : GW	N/A	⑦	boolean	*	wx:if	page data	none	button tapping	①	✓	✓	✓	✓
WeChat	C20	GS : EW : GW	N/A	EW : GS	⑦	*	*	*	execution frequency	coexisting (element)	button tapping	⑥	✗	✗	✗	✓
	C21	WR : NR	N/A	N/A	⑦	*	*	*	execution frequency	none	reloading	①	✗	✗	✓	✓
Alipay	C27	WR : NR	N/A	N/A	⑨	*	*	attribute	variable values	none	button tapping	⑥	✗	✗	✓	✗

SuperApp	Issue	Discrepancies			Data Channel	Data Type	Page Element	Page Position	Observation Oracle	Combination	UI Interaction	Root Cause	Fixed	O ⁻	C ⁻	U ⁻
		EW / WR	GW	GS / NR												
WeChat	P1	 	 	N/A	⑦	*	*	content	execution frequency	none	none	②	✗	✗	✓	✓
	P8	 	 	N/A	⑦	*	*	*	execution order	none	button tapping	⑥	✗	✗	✓	✓
	P9	 	N/A	N/A	①	array	*	attribute	page data	none	button tapping	⑥	✗	✓	✓	✓
Alipay	P10	 	N/A	N/A	③	array	*	attribute	page data	none	button tapping	⑥	✗	✓	✓	✓
	P11	 	N/A	N/A	⑧	*	*	*	variable values	none	button tapping	⑥	✓	✗	✓	✗
	P12	 	N/A	N/A	⑦	*	*	content	execution frequency	none	none	②	✗	✗	✓	✓
TikTok	P18	 	N/A	N/A	③ ⑦	*	*	*	execution frequency	coexisting (all)	none	⑥	✗	✗	✗	✓

Table 2: Known issues of logic discrepancies in cross-layer communications.

Issue	Discrepancies	Data Channel	Data Type	Page Element	Page Position	Observation Oracle	Combination	UI Interaction	Super App (Version)	Reference
K1	 : EW : GS	③	array	*	*	page data	none	none	WeChat (3.4.0)	[4]
K2	 : GW : GS	③	array	*	content	page data	none	none	WeChat (3.4.0)	[8]
K3	 : GW : GS	①	object	*	content	page data	none	none	WeChat (3.4.3)	[7]
K4	 : EW : GS	③	string	*	*	page data	none	none	WeChat (3.4.0)	[9]
K5	 : EW : GS	③	array	*	wx:for	page data	none	none	WeChat (3.4.3)	[5]
K6	 : EW : GS	③	array	*	attribute	page data	none	none	WeChat (3.0.2)	[1]
K7	GW :  	⑨	*	*	attribute	variable values	none	button tapping	WeChat (3.5.2)	[3]
K8	 : EW : GS	⑨	*	*	attribute	variable values	none	button tapping	WeChat (3.5.8)	[6]
K9	GS :  	⑨	*	*	attribute	variable values	none	button tapping	WeChat (3.4.10)	[10]
K10	WR :  	⑨	*	*	attribute	variable values	none	button tapping	Alipay (2.9.73)	[13]
K11	 	①	array	*	a:for	page data	none	none	TikTok (1.57.0)	[2]

We have corrected the discrepancy in the reported results, with all modified table entries highlighted in red. In particular, we now explicitly distinguish between partially fixed cases (denoted as ) and fully fixed cases (denoted as 13

Table 3: Zero-day logic discrepancies in cross-layer communications of WeChat.

Issue	Discrepancies			Data Channel	Data Type	Page Element	Page Position	Observation Oracle	Combination	UI Interaction	Root Cause	Fixed	O ⁻	C ⁻	U ⁻
															
C1	GW,EW,GS	EW,GW	N/A	⑦	*	*	*	execution frequency	none	button tapping	①	✗	✗	✓	✓
C2	GW,EW,GS	EW,GW	N/A	⑥	boolean	*	wx:if	page data	none	button tapping	①	✗	✓	✓	✓
C3	GW,EW,GS	EW,GW	N/A	⑦	boolean	*	wx:if	page data	none	button tapping	①	✗	✓	✓	✓
C4	GS,EW,GW	N/A	EW,GS	⑥	*	*	*	execution frequency	none	reloading	①	✗	✗	✓	✗
C5	GS,EW,GW	N/A	EW,GS	⑦	*	*	*	execution frequency	none	reloading	①	✗	✗	✓	✗
C6	EW,GW,GS	EW,GW	EW,GS	③	array	*	content	page data	none	button tapping	③	✗	✓	✓	✓
C7	EW,GW,GS	EW,GW	EW,GS	③	array	*	attribute	page data	none	button tapping	③	✗	✓	✓	✓
C8	EW,GW,GS	EW,GW	EW,GS	⑦	array	*	attribute	variable values	none	button tapping	③	✗	✓	✓	✓
C9	GS,EW,GW	N/A	EW,GS	②	function	*	*	execution frequency	none	none	③	✗	✓	✓	✓
C10	EW,GW,GS	EW,GW	EW,GS	①	array	*	wx:for-item	page data	conflicting	none	④	✗	✓	✗	✓
C11	EW,GW,GS	EW,GW	EW,GS	②	array	*	wx:for-item	page data	conflicting	none	④	✗	✓	✗	✓
C12	EW,GW,GS	EW,GW	EW,GS	③	array	*	wx:for-item	page data	conflicting	button tapping	④	✗	✓	✗	✓
C13	EW,GW,GS	EW,GW	EW,GS	⑤	array	*	wx:for-item	page data	conflicting	none	④	✗	✓	✗	✓
C14	EW,GW,GS	EW,GW	EW,GS	⑥	array	*	wx:for-item	page data	conflicting	none	④	✗	✓	✗	✓
C15	EW,GW,GS	EW,GW	EW,GS	⑦	array	*	wx:for-item	page data	conflicting	button tapping	④	✗	✓	✗	✓
C16	GS,EW,GW	N/A	EW,GS	⑦	*	*	*	execution frequency	coexisting (data)	button tapping	⑤	✗	✗	✗	✓
C17	GS,EW,GW	N/A	EW,GS	⑥ ⑦	*	*	*	execution frequency	coexisting (all)	button tapping	⑤	✗	✗	✗	✓
C18	GS,EW,GW	N/A	EW,GS	③ ⑥	*	*	*	execution frequency	coexisting (all)	button tapping	⑤	✗	✗	✗	✓
C19	GS,EW,GW	N/A	EW,GS	④ ⑥	*	input	*	execution frequency	coexisting (all)	text entering	⑤	✗	✗	✗	✓
C20	GS,EW,GW	N/A	EW,GS	⑦	*	*	*	execution frequency	coexisting (element)	button tapping	⑤	✗	✗	✗	✓

Issue	Discrepancies			Data Channel	Data Type	Page Element	Page Position	Observation Oracle	Combination	UI Interaction	Root Cause	Fixed	O ⁻	C ⁻	U ⁻
	EW	GW	GS												
P1	  	  	N/A	⑥	*	*	content	execution frequency	none	none	②	✗	✗	✓	✓
P2	  	  	N/A	⑦	*	*	content	execution frequency	none	none	②	✗	✗	✓	✓
P3	N/A	  		①	*	*	*	page data	none	none	③	✗	✓	✓	✓
P4	N/A	  		⑤	*	*	*	page data	none	none	③	✗	✓	✓	✓
P5	N/A	N/A		⑨	*	*	attribute	variable values	none	button tapping	③	✗	✗	✓	✗
P6	  	  	N/A	③ ⑥	*	*	*	execution frequency	coexisting (all)	none	⑤	✓	✗	✗	✓
P7	  	  	N/A	③ ⑦	*	*	*	execution frequency	coexisting (all)	none	⑤	✓	✗	✗	✓
P8	  	  	N/A	⑦	*	*	*	execution order	none	button tapping	⑥	✗	✗	✓	✓

Table 4: Zero-day logic discrepancies in cross-layer communications of Alipay.

Issue	Discrepancies		Data Channel	Data Type	Page Element	Page Position	Observation Oracle	Combination	UI Interaction	Root Cause	Fixed	O ⁻	C ⁻	U ⁻
														
C21	WR, NR		⑥	*	*	*	execution frequency	none	reloading	①	✗	✗	✓	✗
C22	WR, NR		⑦	*	*	*	execution frequency	none	reloading	①	✗	✗	✓	✗
C23	WR, NR		①	function	*	content	variable values	none	none	③	✗	✓	✓	✓
C24	WR, NR		②	function	*	*	variable values	none	none	③	✗	✓	✓	✓
C25	WR, NR		③	function	*	content	variable values	none	button tapping	③	✗	✓	✓	✓
C26	WR, NR		⑤	function	*	content	variable values	none	none	③	✗	✓	✓	✓
C27	WR, NR		⑨	*	*	attribute	variable values	none	button tapping	③	✗	✗	✓	✗

Issue	Discrepancies		Data Channel	Data Type	Page Element	Page Position	Observation Oracle	Combination	UI Interaction	Root Cause	Fixed	O ⁻	C ⁻	U ⁻
														
P9			①	array	*	attribute	page data	none	none	③	✗	✓	✓	✓
P10			③	array	*	attribute	page data	none	button tapping	③	✗	✓	✓	✓
P11			⑧	*	*	*	variable values	none	button tapping	③	✓	✗	✓	✗

Table 5: Zero-day logic discrepancies in cross-layer communications of TikTok.

Issue	Discrepancies		Data Channel	Data Type	Page Element	Page Position	Observation Oracle	Combination	UI Interaction	Root Cause	Fixed	O ⁻	C ⁻	U ⁻
														
P12			⑥	*	*	content	execution frequency	none	none	②	✗	✗	✓	✓
P13			⑦	*	*	content	execution frequency	none	none	②	✗	✗	✓	✓
P14			①	array	*	a:for	page data	none	none	③	✓	✓	✓	✓
P15			③	array	*	a:for	page data	none	none	③	✓	✓	✓	✓
P16			⑤	array	*	a:for	page data	none	none	③	✓	✓	✓	✓
P17			③ ⑥	*	*	*	execution frequency	coexisting (all)	none	⑤	✗	✗	✗	✓
P18			③ ⑦	*	*	*	execution frequency	coexisting (all)	none	⑤	✗	✗	✗	✓